SOCIAL WORK & SUSTAINABILITY · FINAL

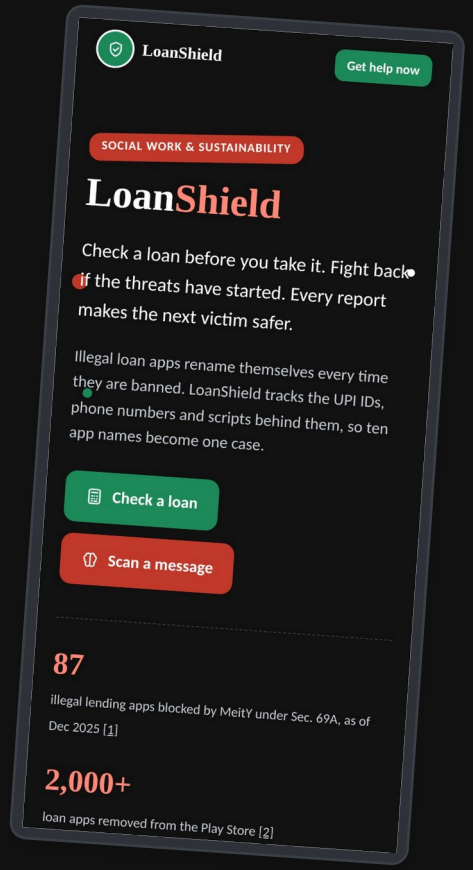
LoanShield

Follow the money, not the app name.

A socio-technical system against illegal loan app harassment in India, built as a working web app and a five-stage community pipeline.

Live app https://claude.ai/artifact/RvuocA8TjtWS3NRL1imJfX

[Team names] · [Course code] · [Institution]



Easy loans. Brutal collection.

Illegal instant-loan apps target low-income, first-time borrowers

Contact-list blackmail

Family, friends and employers get the threats.

Morphed photos

Edited images sent to everyone you know.

Fake legal notices

"FIR", "arrest warrant", "digital arrest".

Hidden cost

Fees cut upfront, full repayment in 7 days.

87

illegal lending apps blocked by MeitY under Sec. 69A (Dec 2025)

2,000+

loan apps removed from the Play Store

67.8%

of cybercrime complaints are online financial fraud (NCRB)

Sources: PIB, Dec 2025; Rajya Sabha Q. 263, Feb 2024; NCRB via Business Standard, Dec 2024.

Four UN goals, all hit by one scam.

1

No Poverty

Traps poor families in debt spirals that cost over 2,000% a year.

8

Decent Work

Threats reach workplaces; victims lose jobs and income.

10

Reduced Inequalities

Those with the least literacy and access are targeted most.

16

Peace & Justice

Victims stay silent; operators escape by renaming apps.

Objectives

What LoanShield sets out to achieve

01

Prevent

See the true cost

Show anyone the real yearly cost of a loan offer, and its rule violations, before they borrow.

02

Detect

Unmask the operator

Link rebranded apps to the operators behind them through UPI IDs, phone numbers and scripts.

03

Protect

Protect without spying

Help victims recognise threats and shield their contacts, without reading private messages.

04

Respond

Make evidence count

Give victims tamper-evident evidence and a complaint that names the rules broken.

05

Reform

Hold systems to account

Publish operators and time-to-action so regulators and app stores can be measured.

06

All stages

Leave no one out

Work in Indian languages and on basic phones, with community volunteers.

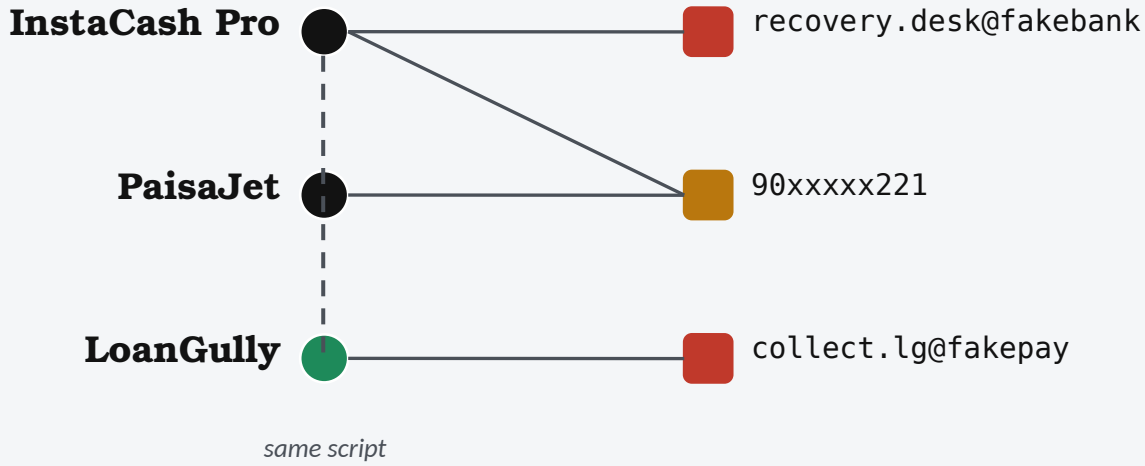
What exists, and the gap it leaves.

Real tools, each seeing part of the picture

Tool	What it does	The gap
Play Store loan policy (Google)	Bans contact and photo access; India needs an RBI-listed app	Misses apps installed from WhatsApp or Telegram links
DLA directory (RBI, July 2025)	Lists apps run by regulated lenders	Says who is legal, not who is harassing you
Sec. 69A blocking (MeitY)	Blocks named apps	One name at a time; operators rebrand
MuleHunter.AI (RBI Innovation Hub)	Flags mule accounts from bank data	Bank-side only; victims' evidence never reaches it
1930, cybercrime.gov.in, Chakshu	Take individual complaints	Each complaint stands alone
LoanShield	Links victim reports by UPI ID, number and script; chains evidence; warns contacts	Feeds linked, hash-verified cases to all of the above

Apps change names. UPI IDs don't.

The insight our solution is built on

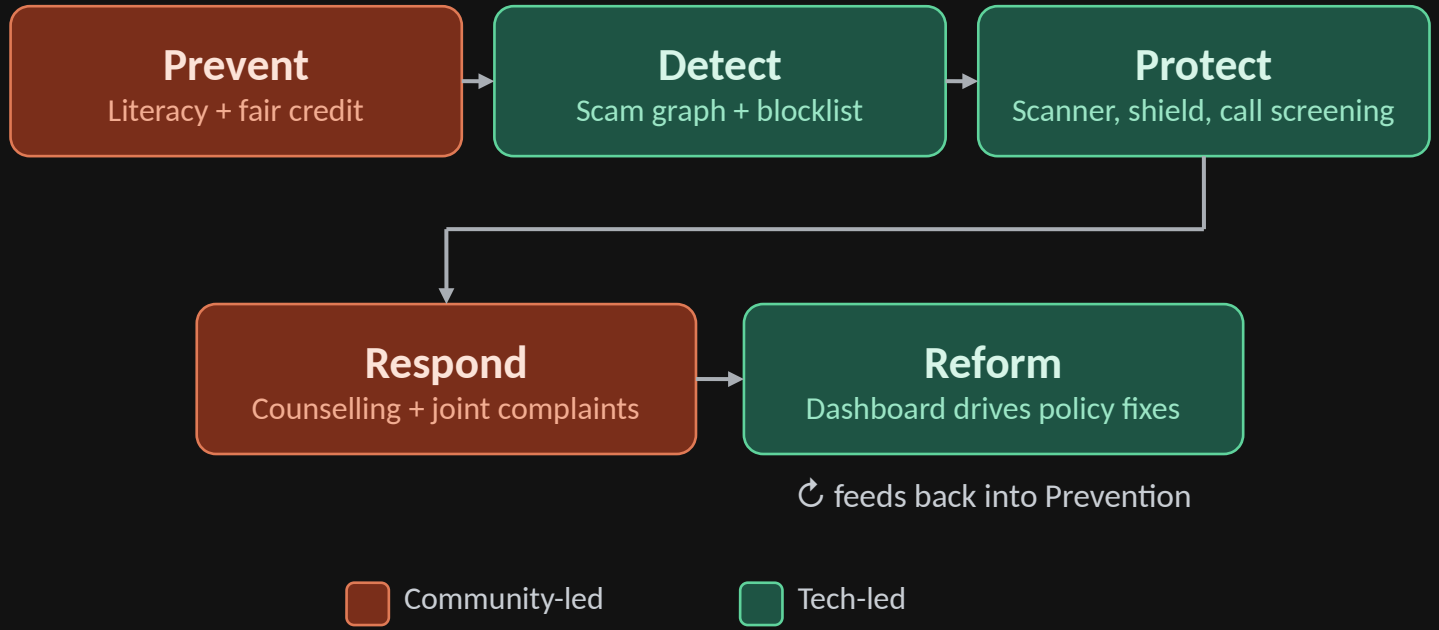


- Same UPI ID
- Same phone number
- Same script (≥ 50%)
- Public at 3 people

Three app names → one operator → one case for the cyber cell

The LoanShield pipeline

Five stages, with a feedback loop back to prevention



Each stage pairs a tool with the people who make it work. The next five slides take one stage each.

STAGE 1 OF 5 · COMMUNITY-LED

Prevent

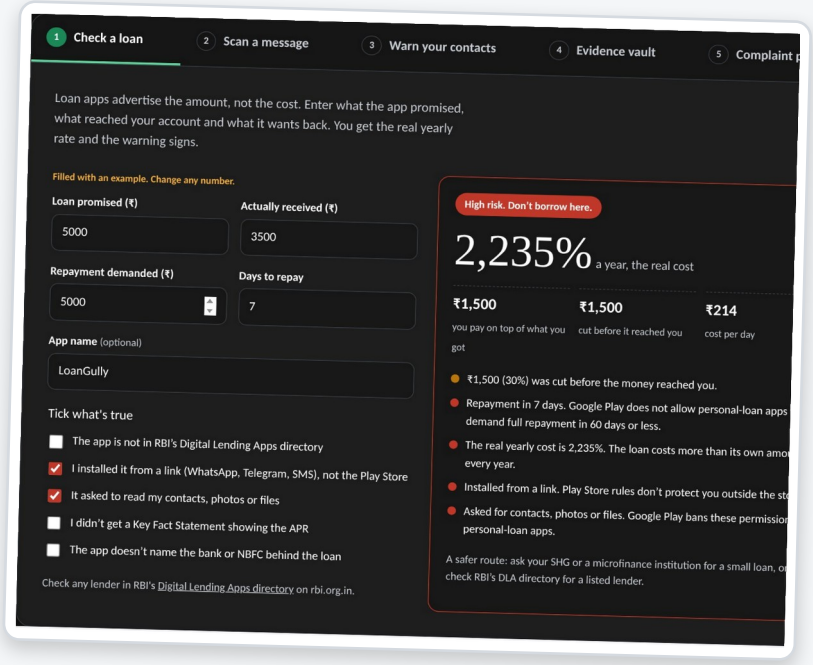
People borrow from scam apps because they need cash fast and cannot see the true cost.

TECH

- Loan Truth Check: ₹5,000 promised, ₹3,500 received, ₹5,000 back in 7 days = 2,235% a year
- Checks RBI and Play Store rules: DLA listing, banned permissions, 60-day minimum, Key Fact Statement
- WhatsApp or IVR lookup of RBI's DLA directory for basic phones

PEOPLE

- SHGs and microfinance institutions offer a fair small loan instead
- Student volunteers run Truth Check camps at SHG meetings



SDG 1.4 SDG 1.5 SDG 8.10

STAGE 2 OF 5 · TECH-LED

Detect

Bans target app names, the cheapest thing a scammer can change.

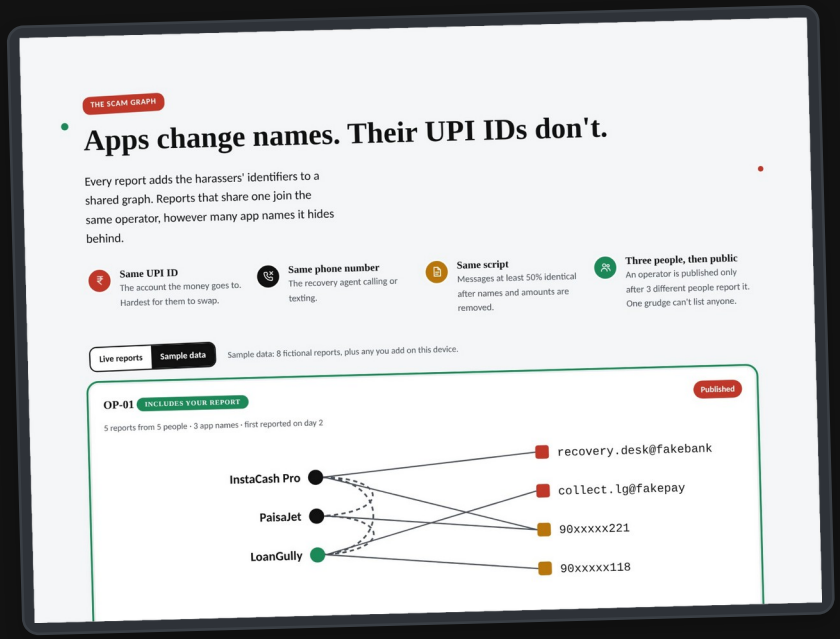
TECH

- Scam graph: union-find links reports that share a UPI ID, a phone number, or a script ≥ 50% similar (Jaccard on 3-word shingles)
- An operator is published only after 3 different people report it
- Live and shared: every report reaches everyone with the link

PEOPLE

- Victims add reports with consent; only identifiers are shared
- App stores and telecoms act on whole operators

SDG 16.4



STAGE 3 OF 5 · TECH-LED

Protect

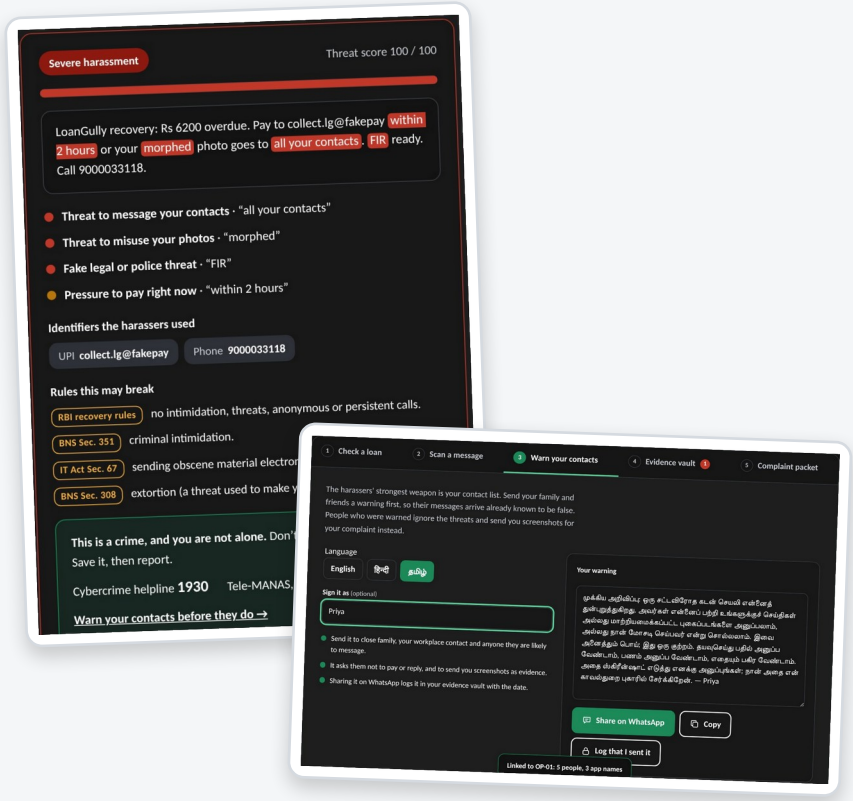
Victims panic and pay more, and their contacts get blackmailed.

TECH

- Share-in threat scanner: 7 threat types in English and Hinglish; pulls out UPI IDs and numbers
- Optional AI second opinion for Tamil, Telugu or any language
- Contact Shield: a warning to your contacts in English, Hindi or Tamil, sent before the harassers reach them

PEOPLE

- Severe threats show 1930 and Tele-MANAS 14416
- NGO volunteers call back within a day



SDG 10.2 **SDG 16.3**

STAGE 4 OF 5 · COMMUNITY-LED

Respond

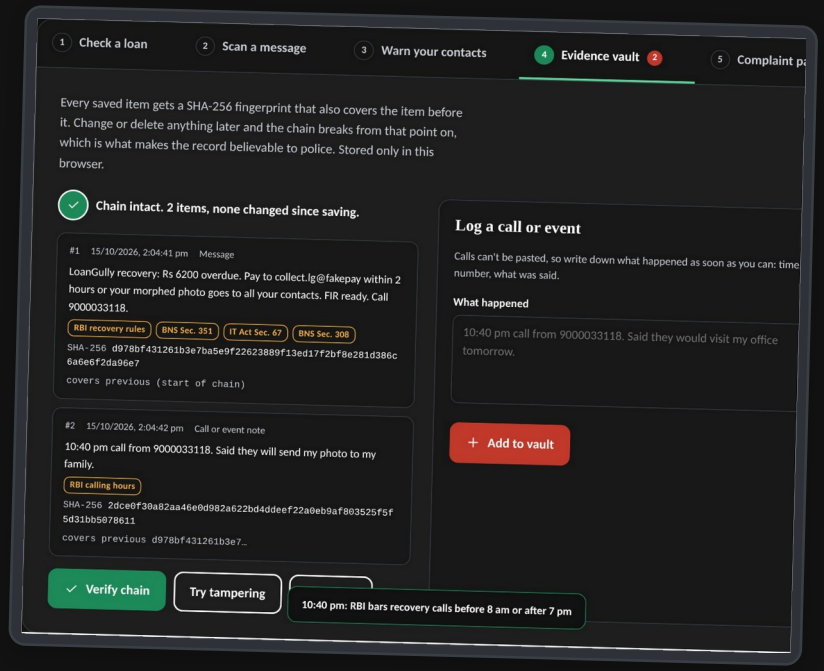
Victims don't report, and the evidence they have is scattered and easy to dispute.

TECH

- Hash-chained vault: each item's SHA-256 also covers the item before it, so later edits are detectable
- Rights map: names the RBI rule, BNS or IT Act section each message may break; flags calls outside 8 am to 7 pm
- Complaint packet for 1930 or cybercrime.gov.in; never filed automatically

PEOPLE

- Law-college legal-aid clinics help victims file and follow up
- One bundled case per operator goes to the cyber cell



SDG 16.3

STAGE 5 OF 5 · TECH-LED

Reform

Enforcement happens out of view, and nobody measures how long a ban takes.



TECH

- Public dashboard: reports, app names, operators published, app names per operator
- Median days from the first report to publishing
- Regional hotspots from report data

What gets counted in public gets fixed.

The dashboard turns many quiet complaints into one public number regulators can be asked about.

PEOPLE

- Students publish a quarterly brief for RBI, MeitY and Google Play
- Hotspots decide where the next Prevent camps run: the loop closes

SDG 16.6

Seven things only LoanShield does

Our novelty: three are new in this final version

Money trail

Links rebranded apps by UPI ID, number and script, so a ban can target the operator.

3-person rule

Nothing is published until 3 different people report it. One grudge can't list anyone.

NEW

Live shared graph

Each report, from anyone with the link, immediately protects the next victim.

Evidence chain

A SHA-256 hash chain shows nothing was edited after saving.

NEW

Rights map

Names the RBI rule, BNS or IT Act section each message may break; flags late-night calls.

NEW

Contact Shield

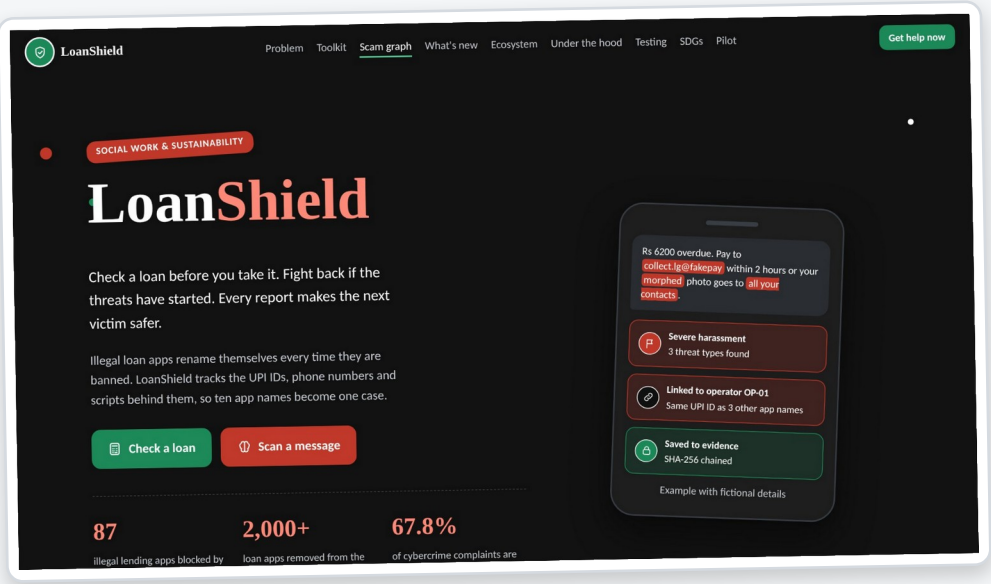
Warns family and colleagues first, in English, Hindi or Tamil, so the blackmail loses its power.

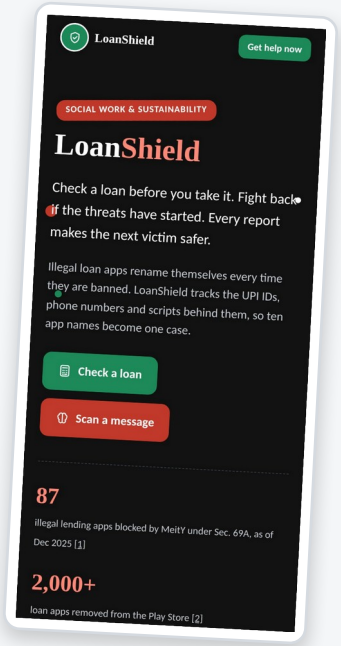
Truth Check

Shows the real yearly cost and rule violations before anyone borrows.

The working product

Live, deployed, and open from one link



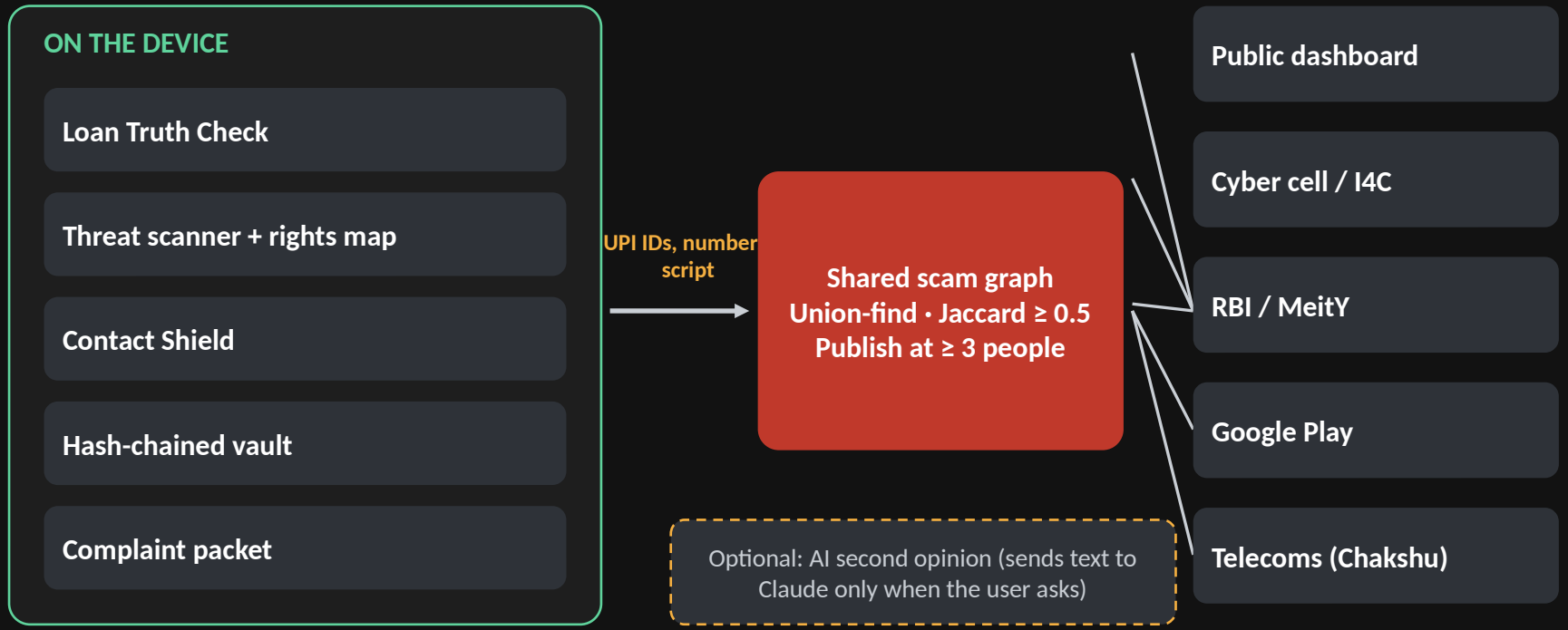


- 5 working tools: Check, Scan, Warn contacts, Evidence vault, Complaint packet
- Live scam graph shared by everyone with the link
- Testing page re-runs the scanner on 50 messages
- Works on phones; nothing typed leaves the browser unless you report

Open it: https://claude.ai/artifact/RvuocA8TjtWS3NRL1imJfX

What runs where

Messages stay on the device. Only the harassers' identifiers leave, and only when you report.



No READ_SMS, READ_CALL_LOG or READ_CONTACTS permission. Designed for the DPDP Act and Rules 2025.

Tech stack

What the live prototype uses, and what the Android app will use

Layer	Live prototype (now)	Android app (Phase 2)
Interface	HTML, CSS, JavaScript; single page; works on phones	Kotlin app with a Share target
Threat detection	Weighted word list, English + Hinglish, 7 threat types; optional Claude second opinion	TensorFlow Lite classifier trained on consented pilot data
Screenshots	—	Google ML Kit text recognition (OCR)
Evidence	SHA-256 hash chain (Web Crypto API), stored in the browser	SQLCipher encrypted database
Scam graph	Union-find + Jaccard similarity on a shared database (Claude Artifacts)	Firebase or a small REST API
Calls	—	Android CallScreeningService
Hosting	Permanent link on claude.ai	Play Store, plus a WhatsApp / IVR line

Does the scanner work?

50 test messages: 25 harassing, 25 ordinary. Re-run live on the website.

100%
harassment caught (recall)

86%
flags that were right (precision)

0.93
F1 score

	Flagged	Not flagged
Harassment	25	0
Ordinary	4	21

Also tested in code: the yearly-rate formula, tamper detection in the hash chain, operator linking, rule mapping and the 8 am to 7 pm check.

The 4 false alarms

- "Please inform your boss."
- "Call your relatives in Madurai today."
- "Arrest of the gang... in today's paper."
- "The cyber cell workshop is at 11am."

One cue with no threat around it. We accept this: people share only messages that worry them, and a missed threat costs more than a false alarm.

Each feature answers an SDG target

The same four goals, mapped to UN targets and the number we track

Target	Meaning	LoanShield feature	We measure
SDG 1.4	Equal access to financial services	Truth Check + SHG/MFI referral	People who check before borrowing
SDG 1.5	Resilience to shocks	Flags upfront cuts, 7-day tenures	Loans declined after a red flag
SDG 8.10	Sound financial institutions	Checks the RBI DLA list	Lookups ending at a listed lender
SDG 10.2	Inclusion of all	Hinglish, Tamil, basic phones, volunteers	Users on basic phones or in Indian languages
SDG 16.3	Equal access to justice	Hash chain, rights map, packet	Complaints filed with a packet
SDG 16.4	Reduce illicit financial flows	Operators linked by UPI ID	UPI IDs reported per operator
SDG 16.6	Accountable institutions	Public dashboard	Median days to action

Challenges, and how we answer them

Apps rebrand in weeks

Link by UPI ID, phone and script, not by name.

Privacy vs. evidence

Android blocks SMS reading and WhatsApp is unreadable, so we use Share-in, with consent.

False or malicious reports

Publish only at 3 reporters; mask phone numbers; share identifiers, never names.

Many Indian languages

Hinglish word list now, AI second opinion, community translation in the pilot.

Shame and fear of reporting

Contact Shield, volunteer support, and a packet that is ready to file.

No public dataset of threats

Team-written test set now; consented labels from pilot users next.

Limitations

What the current version can’t do yet

Keyword scanner

Misses new wording; flags single cues out of context (86% precision on our set).

Test set not from the field

We wrote the 50 messages ourselves; real accuracy is unmeasured until the pilot.

Evidence lives in the browser

Clearing browser data deletes it, and it isn’t encrypted at rest in the prototype.

Shared graph needs sign-in

Visitors outside the project can read the graph but can’t add to it yet.

Legal mapping is indicative

The rights map is a pointer, not legal advice; legal aid must confirm it.

Android features are designed, not built

Call screening and screenshot OCR wait for Phase 2; the packet is never auto-filed.

Future scope

From prototype to a public service

Android app

Share-in scanning, call screening and ML Kit OCR of loan-offer screenshots.

Trained model

TensorFlow Lite classifier on consented pilot data, in Tamil, Telugu and Bengali.

WhatsApp and IVR line

DLA lookup and Truth Check by voice or text for basic phones.

Graph API for enforcement

Operator data for cyber cells and I4C; cross-check UPI IDs against bank mule-account flags.

Fair-credit partners

SHG federations and MFIs offering small loans at the point of need.

Scale and open data

More districts; an anonymised script dataset for researchers.

90-day pilot in one district

Partners: an SHG federation, a law-college legal-aid clinic, the district cyber cell

WEEKS 1–4

Teach and check

- Truth Check camps at SHG meetings
- Translate the word list
- Agree the cyber-cell route

WEEKS 5–8

Scan and save

- Volunteers help victims save evidence
- Users mark scans right or wrong
- Reports enter the live graph

WEEKS 9–12

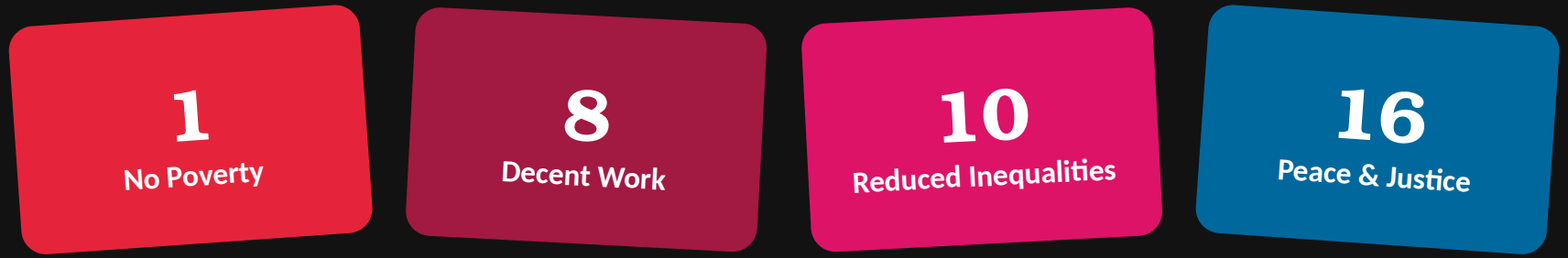
Bundle and report

- Bundled cases to the cyber cell
- First public dashboard
- Brief for RBI and MeitY

Success measure by day 90	Target
People who check a loan before borrowing	200
Scans users mark correct	85%+
Operators linked across 2+ app names	3+
Complaints filed with a packet	25
Users who felt less alone (exit survey)	70%+

Safer loans.
Stronger people.

Follow the money, not the app name. Every report makes the next victim safer.



Try it: https://claude.ai/artifact/RvuocA8TjtWS3NRL1imJfX

Need help now? Cybercrime helpline 1930 · cybercrime.gov.in · Tele-MANAS 14416